

Staged Machine Learning Prediction of Postoperative Complications After Esophagectomy

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Background

Esophagectomy for esophageal cancer has one of the highest postoperative complication rates in surgery (57.6%) [1], yet existing preoperative risk models relying on demographics and comorbidities alone show only moderate discrimination. This study investigates whether staged machine learning, progressively incorporating clinical, tumour, surgical, and ICF functional predictors extracted from clinical notes, can improve preoperative risk stratification before esophagectomy.

Research Questions

Research Question: Can staged ML improve preoperative complication prediction, and which information stage drives the largest gain?

- SQ 1 Does each predictor stage improve discrimination?
SQ 2 Which predictors drive model output? (SHAP analysis)
SQ 3 Can the ML model outperform expert physiotherapy risk classification?
SQ 4 Do preoperative ICF scores add predictive value?
SQ 5 How does patient functioning change at 6 months post-surgery?

Methods

Data — Surgical: Retrospective cohort, Amsterdam UMC (DUCA registry), 2016–2025; N = 1,181 esophagectomy patients; primary outcome: any complication within 30 days (57.6%); secondary: pulmonary, cardiac, anastomotic leak, infection.
Data — ICF: A-Proof NLP [3] extracts 17 ICF functional domains (scale 0–4) from all clinical notes; pre-op baseline = last note ≤30d before surgery (n = 578 patients); post-op = last note ≤180d after surgery (n = 585 patients).
ML pipeline: Four staged models (Stages 1–4) progressively incorporate predictor groups (see diagram); algorithms: LASSO LR · Random Forest (300 trees) · XGBoost; 5-fold stratified CV; evaluated on AUC-ROC, balanced accuracy, Brier score.

Stage 1	Demographics & Fitness <i>Age · Sex · BMI · ASA score</i>
Stage 2	Comorbidities & Lifestyle <i>CCI · Smoking · Weight loss · Cardiac</i>
Stage 3	Tumour & Neoadjuvant <i>cT/cN stage · Histology · Treatment</i>
Stage 4	Surgical Factors ★ <i>Anastomosis · Approach · Duration · Blood loss</i>

SHAP: Global feature importance bar chart + beeswarm plot generated for the best model (Stage 4 Random Forest); unified Shapley framework [5] quantifies each predictor's contribution to individual risk predictions, revealing which clinical factors dominate.
ICF exploration: Pre- vs post-operative ICF functional domain scores compared using Wilcoxon signed-rank test (paired, non-parametric) to quantify the trajectory of functional recovery after esophagectomy (n = 456-578 patient pairs per domain, depending on domain coverage in clinical notes).

Model Performance: Stage Trajectory & ROC

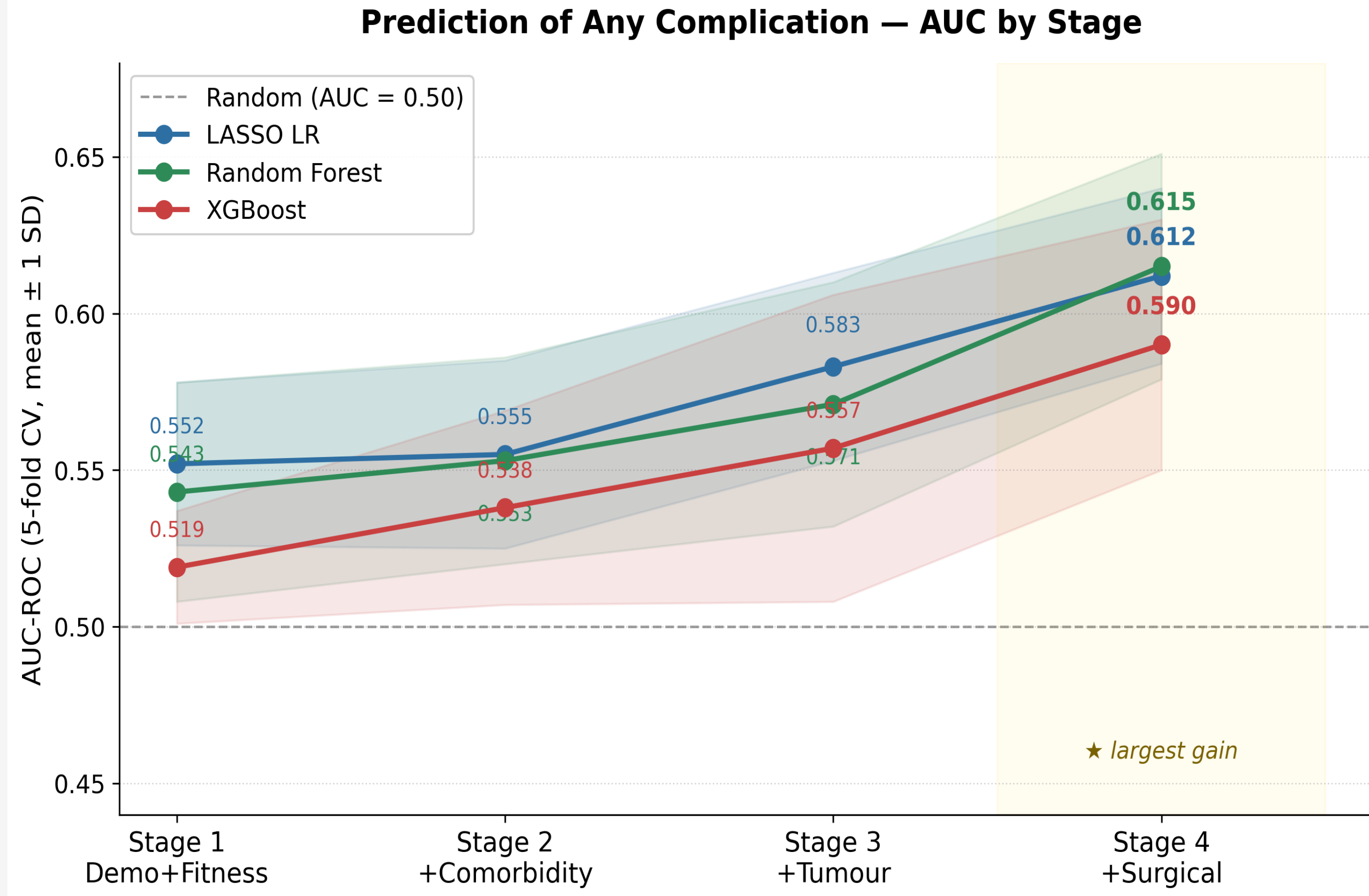


Fig 1. AUC-ROC by stage - mean ±1 SD (5-fold CV). Surgical factors (Stage 4) produce the largest gain across all three algorithms.

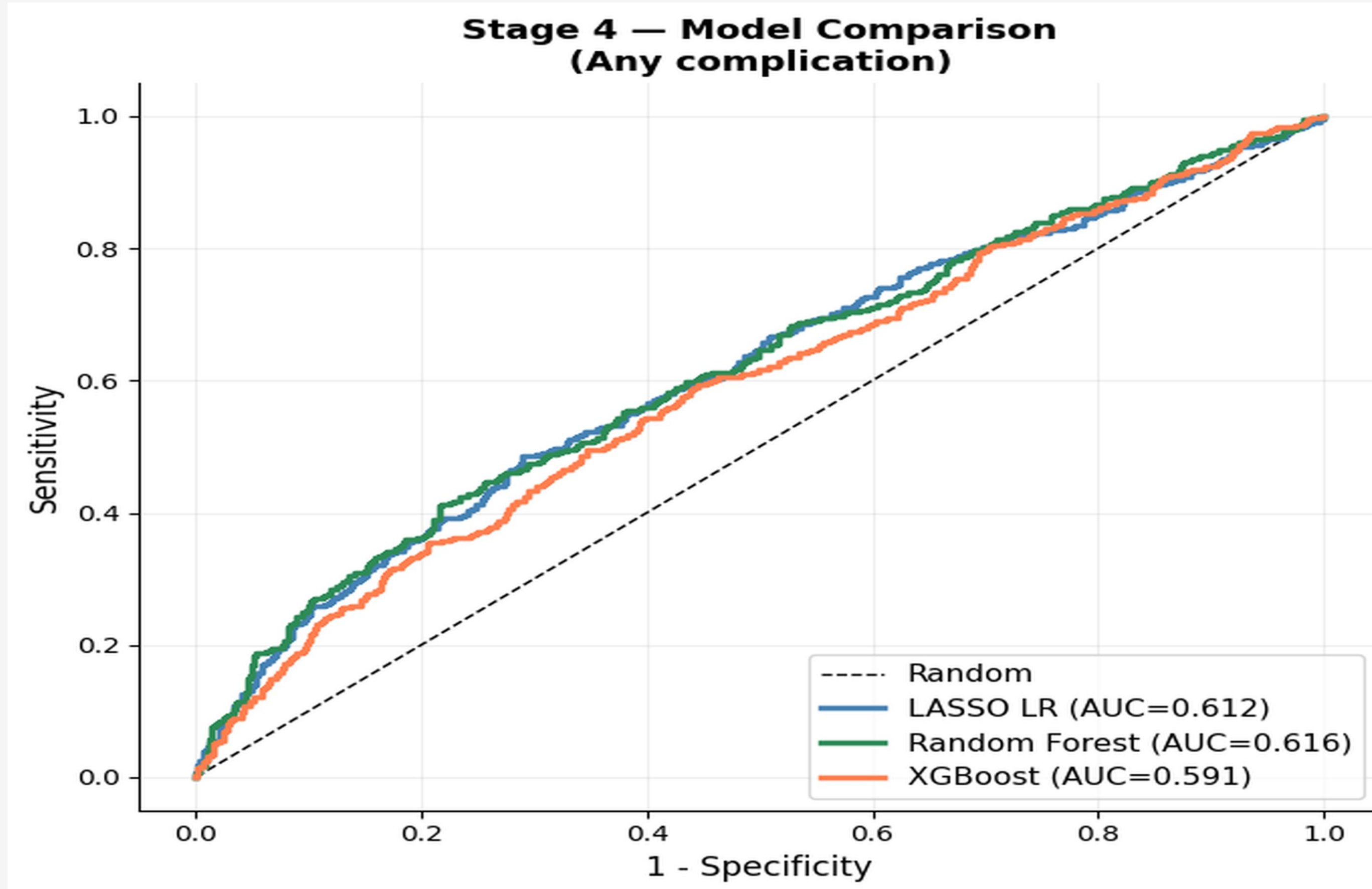


Fig 2. ROC curves at Stage 4 (out-of-fold). LASSO LR and RF nearly indistinguishable; XGBoost marginally lower.

Results: Staged Prediction [SQ 1 & 2]

- Best model: RF Stage 4 - AUC = 0.615, balanced accuracy = 0.585
- Surgical factors (Stage 4) produce the largest gain; AUC ceiling set by predictor availability, not algorithm (all 3 converge, ΔAUC < 0.03)
- Stage 5 sensitivity (FEV1%, n=532): ΔAUC up to +0.020 for RF

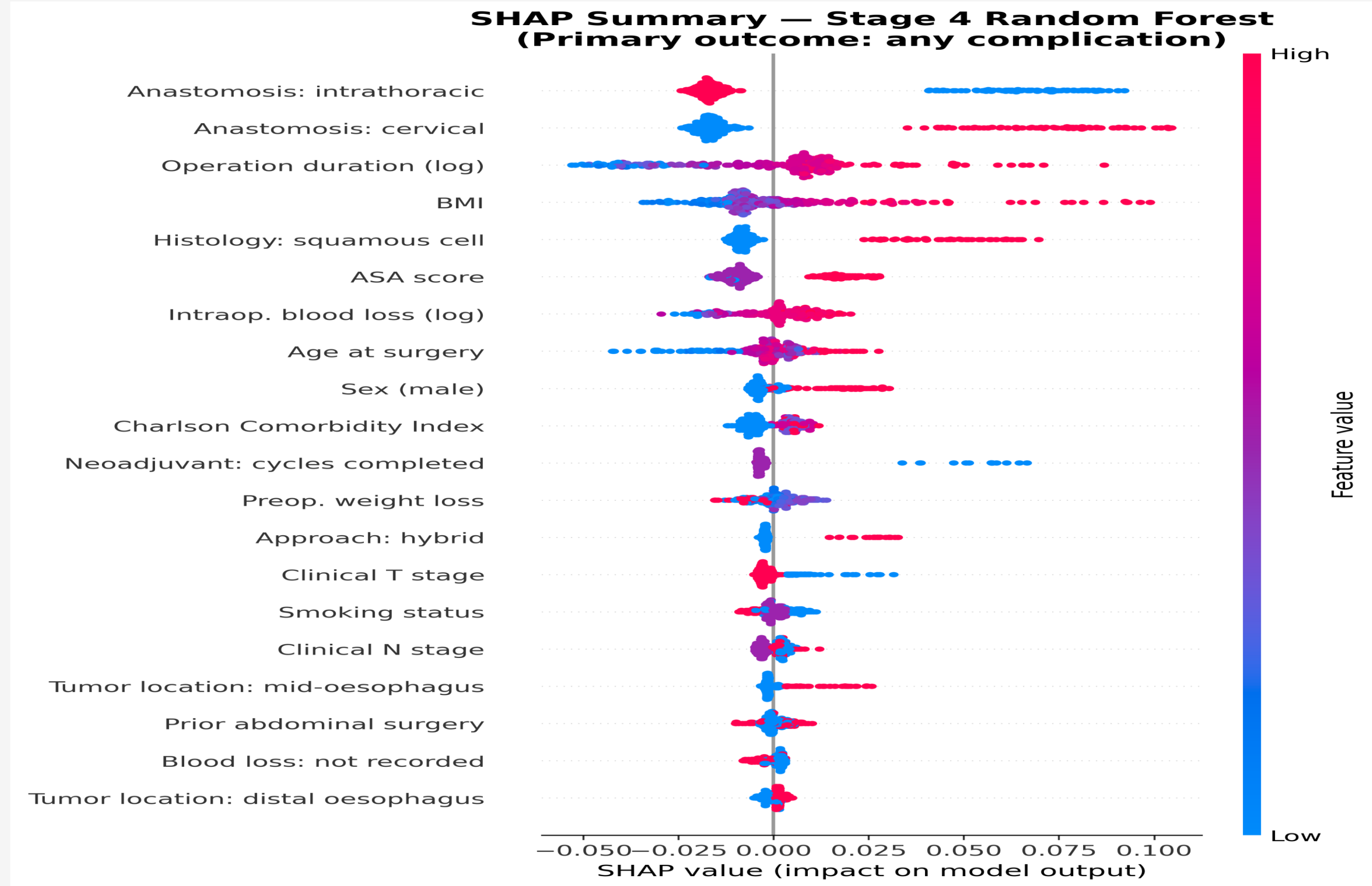
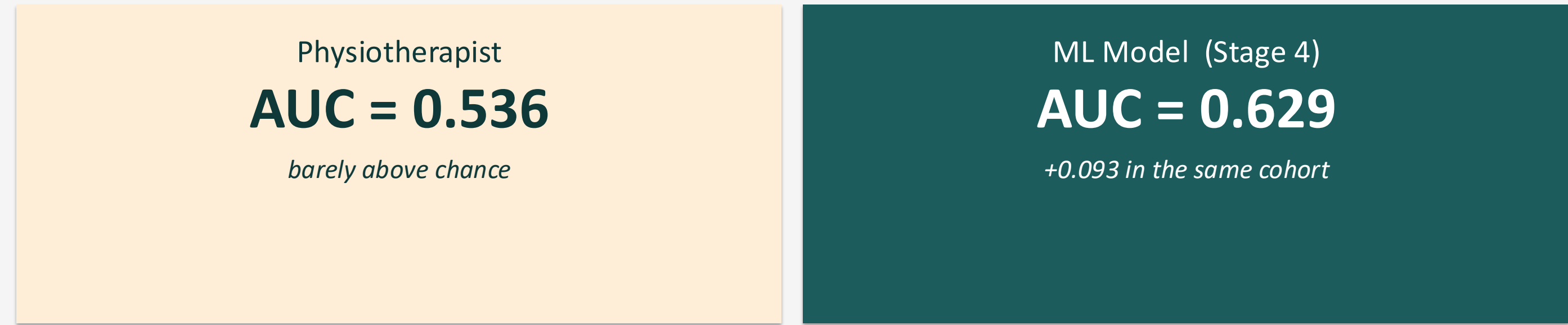


Fig 3. Top predictors by mean |SHAP| - Stage 4 Random Forest. Anastomosis location and operation duration dominate.

Physiotherapist vs. ML Model [SQ 3]

N = 517 patients received a pre-operative qualitative risk assessment (high / low / none) by the project physiotherapist (Dr. E. Geleijn, Amsterdam UMC), based on clinical consultation, physical function, and medical history. This offers a clinically grounded benchmark: can the ML model, trained purely on structured pre-operative data, match or exceed expert physiotherapist judgment? The Stage 4 RF model was applied to the exact same cohort and outcome. AUC-ROC compared head-to-head:



Results: ICF Functional Analysis [SQ 4 & 5]

- 11 of 17 ICF domains changed significantly at 6 months (p < 0.05; pre-op notes available for 48% of cohort; scale 0–4, positive Δ = improvement):
- Most physical domains worsened - exercise tolerance Δ = -0.64 (largest decline), respiration Δ = -0.25, energy Δ = -0.23, pain Δ = -0.18
 - Eating improved most (Δ = +0.66) - resolution of tumour-related obstructive dysphagia
 - Cognition (Δ = +0.51) and sleep (Δ = +0.24) also improved
 - ICF predictive value (n=132): ΔAUC = -0.023 (LR), -0.012 (RF) - underpowered

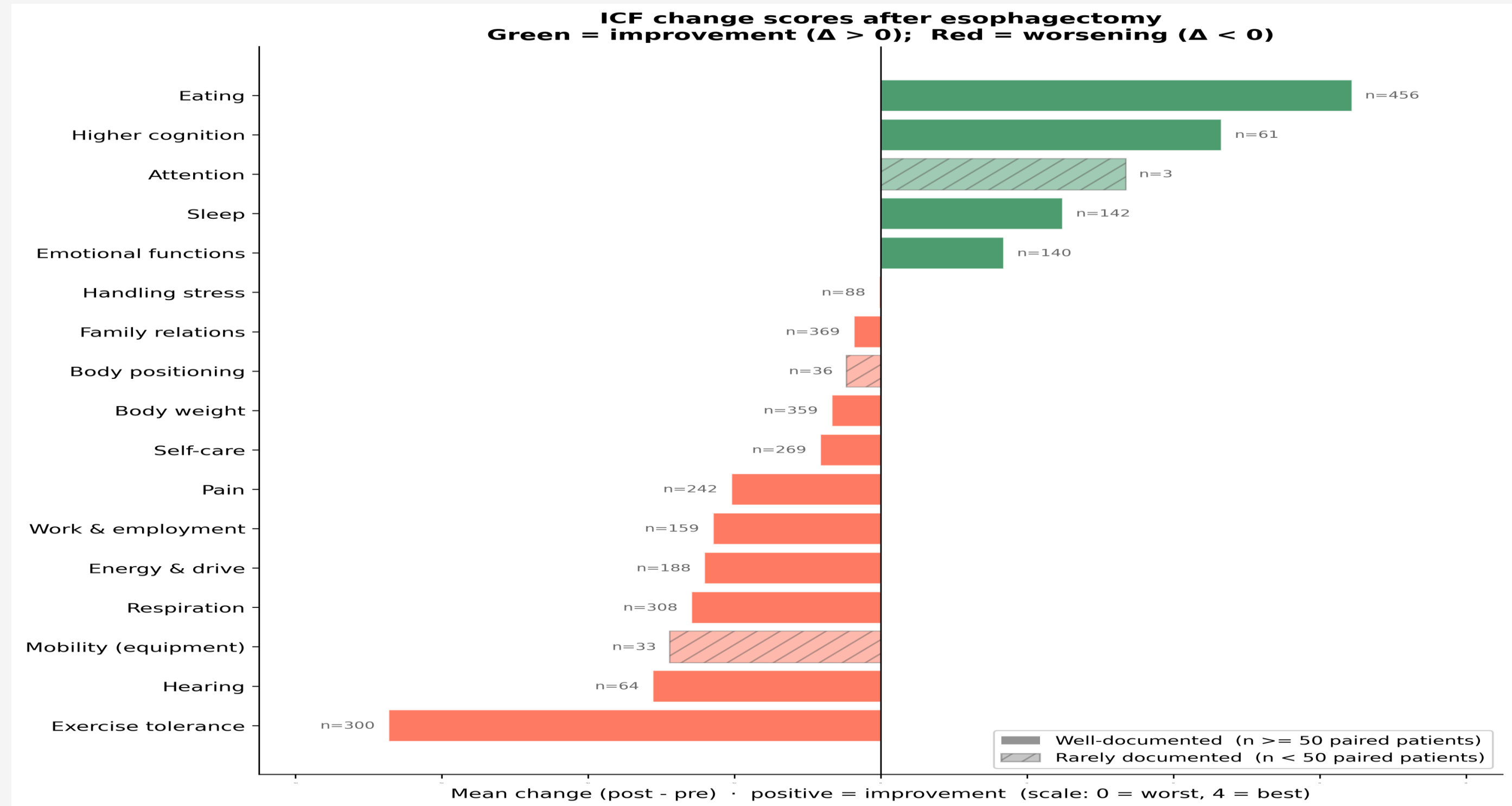


Fig 4. Mean ICF change per domain (pos = improvement, scale 0-4). Green = improvement; red = worsening. n shown per bar.

Key Findings, Conclusion & Limitations

- KEY FINDINGS:**
- RF Stage 4: best model (AUC 0.615); surgical factors drive the largest stage gain (+0.04 AUC)
 - ML outperforms expert physiotherapy classification (AUC 0.629 vs 0.536, N=517); physio missed 56% of complications at high-risk threshold
 - ICF: eating improved most (D = +0.66, dysphagia resolved); exercise tolerance worsened most (D = -0.64)
 - AUC ceiling (~0.62) is set by predictor availability, not by algorithmic choice
- LIMITATIONS:**
- Single-centre retrospective cohort; external validity unknown
 - ICF pre-op coverage 48% (non-random; selection bias toward impaired patients)
 - ICF predictive analysis underpowered (n=132 complete cases)
 - Complication subtypes largely unpredictable preoperatively (AUC 0.49-0.57)

References

- [1] Sung et al., CA Cancer J Clin 71, 209 (2021).
- [2] González Santos et al., Front. Anesthesiol. 3 (2024).
- [3] Guinan et al., BMC Cancer 19, 682 (2019).
- [4] Lundberg & Lee, Adv. Neural Inf. Process. Syst. 30 (2017).
- [5] Dindo et al., Ann. Surg. 240, 205 (2004).